

ICSA 2026 SHORT COURSE

Lecture 2: Domain adaptation via divergence notions

Statistical Foundations of Transfer Learning

YANG FENG¹ YE TIAN²

¹NYU

²Yale University

Overview

§2.1: No target data

Numerical illustration: shift and excess risk

§2.2: Few target data + many source data

§2.3: Posterior drift (preview; returns in L4)

§2.4: Hardness

§2.5: Beyond divergence

References

§2.1: No target data

Setup: source-only learning

Recall L1's running example: an Alzheimer's classifier trained on research-study MRIs (the source), to be deployed at a community hospital (the target) — but with no target labels yet. How much does the domain gap cost?

Start with noiseless binary classification. Define

- Target: $X \sim \mathbb{P}^{(0)}$, $Y = f^{(0)}(X) \in \{0, 1\}$. Source: $X \sim \mathbb{P}^{(1)}$, $Y = f^{(1)}(X)$.
- Concept drift: both $\mathbb{P}^{(0)} \neq \mathbb{P}^{(1)}$ and $f^{(0)} \neq f^{(1)}$ allowed.
- Risk $R^{(k)}(h) = \mathbb{E}_{X \sim \mathbb{P}^{(k)}} [\mathbf{1}\{h(X) \neq f^{(k)}(X)\}]$, hypothesis class $\mathcal{H}$.
- Observed: $\{(\mathbf{x}_i^{(1)}, y_i^{(1)})\}_{i=1}^{n_1}$ only; no target labels at all.
- Goal: a classifier $\hat{h}$ with small **target** risk $R^{(0)}(\hat{h})$.

The natural strategy is to use $\hat{h}^{(1)} = \arg \min_{h \in \mathcal{H}} \hat{R}^{(1)}(h)$ and hope for the best. (Noiseless means $\mathbb{P}_{Y|X}^{(k)}$ collapses to a deterministic label rule $f^{(k)}$ with $f^{(k)}(x) = \mathbb{E}[Y | X = x]$; posterior/concept drift then means $f^{(0)} \neq f^{(1)}$ — the same drift L1 and L4 phrase via $\mathbb{P}_{Y|X}^{(0)}$.)

Decomposing the target risk

Add and subtract $R^{(1)}(\hat{h})$ and use $R^{(k)}(h_1, h_2) := \mathbb{E}_{X \sim \mathbb{P}^{(k)}} \mathbf{1}\{h_1(X) \neq h_2(X)\}$: (Here $R^{(k)}(h) = R^{(k)}(h, f^{(k)})$ since labels = $f^{(k)}(X)$ on domain k .)

$$\begin{aligned} R^{(0)}(\hat{h}) &= R^{(1)}(\hat{h}) + [R^{(0)}(\hat{h}, f^{(0)}) - R^{(1)}(\hat{h}, f^{(1)})] \\ &\leq R^{(1)}(\hat{h}) + \underbrace{|R^{(0)}(\hat{h}, f^{(0)}) - R^{(1)}(\hat{h}, f^{(0)})|}_{\text{covariate shift}} + \underbrace{|R^{(1)}(\hat{h}, f^{(0)}) - R^{(1)}(\hat{h}, f^{(1)})|}_{\text{posterior drift}}. \end{aligned}$$

(These two are error contributions — both present at once under concept drift — not the mutually-exclusive regimes named in L1.)

Each bracketed piece is now bounded by a divergence (*next slide*).

Bounding each piece

Take $A = \{\hat{h} \neq f^{(0)}\}$. Each bracketed term collapses to a familiar divergence:

- **Covariate-shift term:**

$$|\mathbb{E}_{\mathbb{P}^{(0)}} \mathbf{1}\{\hat{h} \neq f^{(0)}\} - \mathbb{E}_{\mathbb{P}^{(1)}} \mathbf{1}\{\hat{h} \neq f^{(0)}\}| \leq \sup_A |\mathbb{P}^{(0)}(A) - \mathbb{P}^{(1)}(A)| = d_{\text{TV}}(\mathbb{P}^{(0)}, \mathbb{P}^{(1)}).$$

- **Posterior-drift term:**

$$|R^{(1)}(\hat{h}, f^{(0)}) - R^{(1)}(\hat{h}, f^{(1)})| \leq \mathbb{E}_{X \sim \mathbb{P}^{(1)}} |f^{(0)}(X) - f^{(1)}(X)|.$$

The first bound (in the spirit of Ben-David et al.)

Theorem 2.1.1 (A TV-based decomposition, cf. Ben-David et al., 2006, 2010a)

If $\mathcal{H}$ is a VC class, then with high probability,

$$R^{(0)}(\hat{h}) \leq \underbrace{\min_h R^{(0)}(h)}_{\text{oracle}} + \underbrace{2C \sqrt{\frac{\text{VC}(\mathcal{H})}{n_1}}}_{\text{learning cost}} + \underbrace{2 \mathbb{E}_{X \sim \mathbb{P}^{(1)}} |f^{(1)}(X) - f^{(0)}(X)|}_{\text{posterior drift}} + \underbrace{2 d_{\text{TV}}(\mathbb{P}^{(0)}, \mathbb{P}^{(1)})}_{\text{covariate shift}}.$$

The $2C \sqrt{\text{VC}/n_1}$ term (ERM analysis): $R^{(1)}(\hat{h}) \leq \hat{R}^{(1)}(\hat{h}) + C \sqrt{\text{VC}/n_1}$, then $\hat{R}^{(1)}(\hat{h}) \leq \hat{R}^{(1)}(h^*)$ (ERM), then $\hat{R}^{(1)}(h^*) \leq R^{(1)}(h^*) + C \sqrt{\text{VC}/n_1}$ — uniform convergence at both $\hat{h}$ and h^* gives the two copies.

(d_{TV} = worst-case event-probability gap between source and target, independent of $\mathcal{H}$.)

- Clean “bias-variance + shift” decomposition (the canonical Ben-David constant is the joint-optimal λ^* ; here it is split into explicit drift + TV terms).
- **Drawback:** the shift terms are *independent of $\mathcal{H}$* $\implies$ bound is often loose.

An improvement: use a *learner-aware* divergence

Definition 2.1.2 ($\mathcal{H}\Delta\mathcal{H}$ -divergence, Kifer et al., 2004)

- Symmetric difference class $\mathcal{H}\Delta\mathcal{H} = \{\mathbf{1}\{h(\mathbf{x}) \neq h'(\mathbf{x})\} : h, h' \in \mathcal{H}\}$.
- Set system $I(\mathcal{H}\Delta\mathcal{H}) = \{\{\mathbf{x} : g(\mathbf{x}) = 1\} : g \in \mathcal{H}\Delta\mathcal{H}\}$.
- $d_{\mathcal{H}\Delta\mathcal{H}}(\mathbb{P}, \mathbb{P}') = \sup_{A \in I(\mathcal{H}\Delta\mathcal{H})} |\mathbb{P}(A) - \mathbb{P}'(A)|$.

Intuition: $d_{\mathcal{H}\Delta\mathcal{H}}$ measures how well a pair of hypotheses in $\mathcal{H}$ can detect which domain a point came from — if none can, transfer is safe.

Example: $\mathcal{H} = \text{thresholds on } \mathbb{R} \Rightarrow \mathcal{H}\Delta\mathcal{H} = \text{intervals } [a, b]$; $d_{\mathcal{H}\Delta\mathcal{H}} = \text{largest source-vs-target gap any interval sees}$. Key trick (triangle inequality for 0–1 risk, then $|R^{(1)}(h, h') - R^{(0)}(h, h')| \leq d_{\mathcal{H}\Delta\mathcal{H}}$):

$$\begin{aligned} R^{(0)}(\widehat{h}) &\leq R^{(0)}(h^*) + R^{(0)}(\widehat{h}, h^*) \quad (\text{triangle ineq. for 0-1 loss}) \\ &\leq R^{(0)}(h^*) + R^{(1)}(\widehat{h}, h^*) + d_{\mathcal{H}\Delta\mathcal{H}} \\ &\leq R^{(1)}(\widehat{h}) + [R^{(0)}(h^*) + R^{(1)}(h^*)] + d_{\mathcal{H}\Delta\mathcal{H}}. \quad (\text{triangle: } R^{(1)}(\widehat{h}, h^*) \leq R^{(1)}(\widehat{h}) + R^{(1)}(h^*)) \end{aligned}$$

Picking $h^* = \arg \min_{h \in \mathcal{H}} R^{(0)}(h)$ and bounding $R^{(1)}(\widehat{h})$ via VC gives:

The second bound

Theorem 2.1.3 (The second bound)

With high probability,

$$R^{(0)}(\hat{h}) \leq \underbrace{\min_h R^{(0)}(h)}_{\text{oracle}} + \underbrace{2C\sqrt{\frac{\text{VC}(\mathcal{H})}{n_1}}}_{\text{learning cost}} + \underbrace{2R^{(1)}(h^*)}_{\text{drift + shift}} + \underbrace{d_{\mathcal{H}\Delta\mathcal{H}}(\mathbb{P}^{(0)}, \mathbb{P}^{(1)})}_{\text{learner-aware shift}}.$$

Same $\sqrt{\text{VC}/n_1}$ term; the win is the shift: $2d_{\text{TV}}(\mathcal{H}\text{-blind}) \rightarrow d_{\mathcal{H}\Delta\mathcal{H}}$ (uses only what $\mathcal{H}$ separates), always $\leq 2d_{\text{TV}}$, often $\ll$. **Two improvements over the first bound:**

- $d_{\mathcal{H}\Delta\mathcal{H}}$ uses only sets that $\mathcal{H}$ can detect, not all measurable sets.
- $R^{(1)}(h^*)$ couples drift to the loss ℓ explicitly.

Beyond classification: bounded-loss regression

Allow $\mathcal{Y} \subseteq \mathbb{R}$ bounded, ℓ symmetric & satisfies the triangle inequality.

Definition 2.1.4 (Y -discrepancy, Mansour et al., 2009)

For loss ℓ and class $\mathcal{H}$,

$$\text{disc}_\ell(\mathbb{P}^{(0)}, \mathbb{P}^{(1)}) = \sup_{h, h' \in \mathcal{H}} \left| \mathbb{E}_{\mathbb{P}^{(0)}} \ell(h(X), h'(X)) - \mathbb{E}_{\mathbb{P}^{(1)}} \ell(h(X), h'(X)) \right|.$$

Why this form? For 0-1 loss $\ell = \mathbf{1}\{h \neq h'\}$, disc_ℓ reduces to $d_{\mathcal{H} \Delta \mathcal{H}}$; for squared loss it is the worst-case difference in expected disagreement — learner- and loss-aware.

Theorem 2.1.5 (The third bound, Ben-David et al., 2010a; Mansour et al., 2009)

With high probability,

$$R^{(0)}(\hat{h}) \leq \min_h R^{(0)}(h) + \mathcal{O}\left(\sqrt{\frac{\text{VC}(\mathcal{H})}{n_1}}\right) + 2R^{(1)}(h^*) + \text{disc}_\ell(\mathbb{P}^{(0)}, \mathbb{P}^{(1)}).$$

Summary: no target data

Source-only learning gives a target-risk bound of the shape

$$R^{(0)}(\hat{h}) \leq \text{oracle} + \text{learning cost} + \text{posterior drift} + \text{covariate shift}.$$

- Three increasingly refined bounds: $\text{TV} \rightarrow \mathcal{H} \triangle \mathcal{H} \rightarrow \ell$ -discrepancy.
- All three say: **transfer is only safe when divergence *and* drift are small.**
- None of them tell us *how small*; that's what the hardness section will pin down.

Numerical illustration: shift and excess risk

Numerical illustration: setup

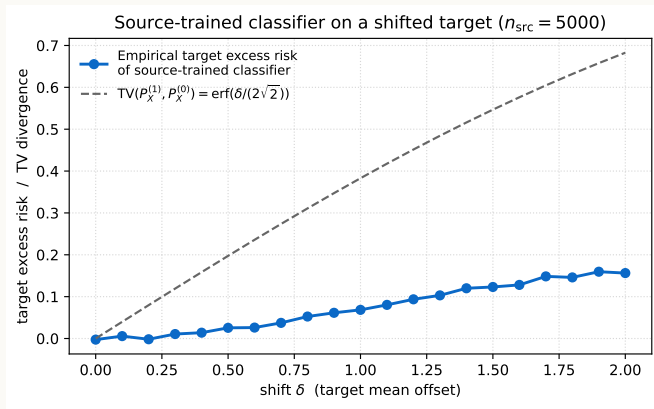
Toy 1-D classification. Source/target Gaussians shifted by δ :

$$P^{(1)} : X | Y=0 \sim N(0, 1), \quad X | Y=1 \sim N(1, 1); \quad P^{(0)} : \text{shift both by } \delta.$$

Train logistic regression on a source sample of size $n_1 = 5000$; evaluate on the target as δ varies. TV of the X -marginals: $d_{\text{TV}}(\mathbb{P}_X^{(1)}, \mathbb{P}_X^{(0)}) \approx \text{erf}(\delta/(2\sqrt{2}))$ (exact per Gaussian component).

Code: `sims/L2_divergence_demo.py` (NumPy, deterministic seed).

Numerical illustration: result



- Empirical target excess risk grows roughly linearly in the shift.
- TV divergence (dashed) is a valid *upper* bound, as theory predicts.
- Empirical curve is well below – VC machinery is loose here.

Numerical takeaway

- Bounds give the right **qualitative** shape, but can be conservative for small shifts.
- This is exactly why we want *learner-aware* divergences (e.g. $d_{\mathcal{H}\Delta\mathcal{H}}$) and *loss-aware* discrepancies: they tighten the bound where the loose TV ignores the geometry of $\mathcal{H}$.
- In the rest of the lecture we will see what happens when target labels *do* arrive in small numbers.

Check-in: when does source-only transfer help?

Suppose your target data are fancy MRI scans and your source data are old MRIs from a different scanner. The two scanners produce features with TV distance $d_{TV} = 0.6$, but the disease-given-features mechanism is unchanged.

Which of the following best describes what the first bound predicts?

- (a) Source-only transfer can reach the oracle target risk with enough source data.
- (b) The bound has a $2d_{TV} = 1.2$ term, so it is *vacuous* (risks $\in [0, 1]$).
- (c) The posterior-drift term dominates because we have no labels.
- (d) The bound is meaningless because we don't know $\mathcal{H}$.

Answer

(b). A large TV term inflates the bound past 1, so although source-only ERM may still help in practice, this particular bound certifies nothing. It motivates learner-aware divergences (§2.1) – and using *some* target data (§2.2).

Check-in: when does source-only transfer help?

Suppose your target data are fancy MRI scans and your source data are old MRIs from a different scanner. The two scanners produce features with TV distance $d_{TV} = 0.6$, but the disease-given-features mechanism is unchanged.

Which of the following best describes what the first bound predicts?

- (a) Source-only transfer can reach the oracle target risk with enough source data.
- (b) The bound has a $2d_{TV} = 1.2$ term, so it is *vacuous* (risks $\in [0, 1]$).
- (c) The posterior-drift term dominates because we have no labels.
- (d) The bound is meaningless because we don't know $\mathcal{H}$.

Answer

(b). A large TV term inflates the bound past 1, so although source-only ERM may still help in practice, this particular bound certifies nothing. It motivates learner-aware divergences (§2.1) – and using *some* target data (§2.2).

§2.2: Few target data + many source data

Setup: a mixture of source and (a few) target labels

Now we also see $\{(\mathbf{x}_i^{(0)}, y_i^{(0)})\}_{i=1}^{n_0}$ with $n_0 \ll n_1$.

- Pure source ERM ignores target structure.
- Pure target ERM throws away n_1 source samples.
- **Compromise:** weighted ERM with mixing parameter $\alpha \in [0, 1]$:

$$\hat{h}_\alpha = \arg \min_{h \in \mathcal{H}} \alpha \hat{R}^{(0)}(h) + (1 - \alpha) \hat{R}^{(1)}(h).$$

Question: how to choose α , and how does the chosen α affect target risk?

Optimal α trades off shift vs. data

Theorem 2.3.1 (Blitzer et al., 2007)

With probability $\geq 1 - \delta$,

$$R^{(0)}(\hat{h}_\alpha) \leq \min_h R^{(0)}(h) + 4\sqrt{\left(\frac{\alpha^2}{m} + \frac{(1-\alpha)^2}{n_1}\right) \text{VC}(\mathcal{H}) \log \frac{2(n_1+m)}{\delta}} + 2(1-\alpha)[\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}],$$

where $m = n_0$ and $\lambda^* = \min_h [R^{(0)}(h) + R^{(1)}(h)]$.

Why $\alpha^2/m + (1-\alpha)^2/n_1$? Each term's variance scales as (weight)²/(sample size): a sample average has variance $\propto 1/n$ and the weight enters squared.

(δ here is the confidence level; unrelated to the mean-shift δ in the numerical illustration.)

- As $\alpha \rightarrow 1$: pure target ERM. Variance scales like $1/n_0$.
- As $\alpha \rightarrow 0$: pure source ERM. Bias scales like $\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}$.
- Optimal α^* shrinks toward 0 as source-target similarity improves.

When does *any* source help? (Intuition)

Differentiating the bound in α and solving: *Differentiate the bound: the variance part gives $\alpha/m - (1 - \alpha)/n_1 = 0 \Rightarrow \alpha = n_0/(n_0 + n_1)$; the bias part $2(1 - \alpha)[\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}]$ pushes α^* up when shift is large.*

$$\alpha^* \approx \min \left\{ 1, \underbrace{\frac{n_0}{n_0 + n_1}}_{\text{variance-optimal}} + \kappa [\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}] \right\}, \quad \kappa = \kappa(n_0, n_1, \text{VC}) > 0.$$

- If $\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}$ is *larger than* the target sampling error, $\alpha^* \rightarrow 1$: ignore the source.
- If shift is *small*, α^* pushes toward $\frac{n_0}{n_0 + n_1}$: treat all data as one big sample, weighted by size.
- This is the rigorous version of “transfer when, and only when, source helps.”

Multiple sources

K sources $\mathbb{P}^{(1)}, \dots, \mathbb{P}^{(K)}$, weights $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_K)$ with $\sum_k \alpha_k = 1$.

Theorem 2.3.2 (Pairwise divergence bound, Ben-David et al., 2010a)

$$R^{(0)}(\hat{h}_{\boldsymbol{\alpha}}) \leq \min_h R^{(0)}(h) + \mathcal{O}\left(\sqrt{(\sum_k \alpha_k^2 / n_k) \text{VC}(\mathcal{H})}\right) + \sum_{k=1}^K \alpha_k [\lambda_k^* + d_{\mathcal{H} \triangle \mathcal{H}}(\mathbb{P}^{(0)}, \mathbb{P}^{(k)})].$$

Theorem 2.3.3 (Combined-divergence bound, Ben-David et al., 2010a)

Tighter when sources jointly concentrate near the target: bias term replaced by $\lambda_{\boldsymbol{\alpha}}^ + d_{\mathcal{H} \triangle \mathcal{H}}(\mathbb{P}^{(0)}, \sum_k \alpha_k \mathbb{P}^{(k)})$.*

Choosing $\boldsymbol{\alpha}$: minimize the empirical surrogate of the bound (a convex QP).

Summary: few target + many source

- Weighted ERM is the natural compromise; α^* is found by optimizing the bound.
- The qualitative rule:

$$\text{use source iff } \underbrace{\sqrt{\text{VC}/n_0}}_{\text{target sampling error}} \gtrsim \underbrace{\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}}}_{\text{shift bias}}.$$

- Multi-source: pairwise vs. combined-divergence bounds; weighting risks vs. hypotheses gives different bias-variance trade-offs.

Check-in: choosing α

You have $n_0 = 200$ target labels and $n_1 = 20,000$ source labels. $VC(\mathcal{H}) = 10$ and you estimate $\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}} \approx 0.05$.

Roughly what α^* does the Blitzer bound suggest?

- (a) $\alpha^* \approx 1$: ignore source; target alone is enough.
- (b) $\alpha^* \approx \frac{n_0}{n_0 + n_1} \approx 0.01$: weight by sample size.
- (c) $\alpha^* \approx 0.5$: half-and-half is always safe.
- (d) $\alpha^* \approx 0$: ignore target labels.

Answer

(b). Target sampling error $\sqrt{10/200} \approx 0.22$ is large compared to shift bias 0.05, so the source helps; the optimal α is dominated by the size-proportional term $n_0/(n_0 + n_1) \approx 0.01$ – almost all weight goes to source.

Check-in: choosing α

You have $n_0 = 200$ target labels and $n_1 = 20,000$ source labels. $VC(\mathcal{H}) = 10$ and you estimate $\lambda^* + d_{\mathcal{H}\Delta\mathcal{H}} \approx 0.05$.

Roughly what α^* does the Blitzer bound suggest?

- (a) $\alpha^* \approx 1$: ignore source; target alone is enough.
- (b) $\alpha^* \approx \frac{n_0}{n_0 + n_1} \approx 0.01$: weight by sample size.
- (c) $\alpha^* \approx 0.5$: half-and-half is always safe.
- (d) $\alpha^* \approx 0$: ignore target labels.

Answer

(b). Target sampling error $\sqrt{10/200} \approx 0.22$ is large compared to shift bias 0.05, so the source helps; the optimal α is dominated by the size-proportional term $n_0/(n_0 + n_1) \approx 0.01$ – almost all weight goes to source.

§2.3: Posterior drift (preview; returns in L4)

Posterior drift: $\mathbb{P}_X^{(0)} = \mathbb{P}_X^{(1)}, f^{(0)} \neq f^{(1)}$

§2.2 mixed source and target under general shift; here we isolate pure posterior drift ($\mathbb{P}_X^{(0)}$ identical across domains).

- Same covariate distribution; only the regression / decision function shifts.
- Setup: target samples $\{(\mathbf{x}_i^{(0)}, y_i^{(0)})\}_{i=1}^{n_0}$, source $\{(\mathbf{x}_i^{(1)}, y_i^{(1)})\}_{i=1}^{n_1}$, $n_0 \ll n_1$.
- Pool with weights $\mathbf{w} \in \Delta^2$:

$$\hat{h}_{\mathbf{w}} = \arg \min_{h \in \mathcal{H}} \frac{w_0}{n_0} \sum_i \ell(h(\mathbf{x}_i^{(0)}), y_i^{(0)}) + \frac{w_1}{n_1} \sum_j \ell(h(\mathbf{x}_j^{(1)}), y_j^{(1)}).$$

Question: how should w_0, w_1 depend on the gap $\Delta = \mathbb{E}_X |f^{(0)}(X) - f^{(1)}(X)|$?

Posterior drift: target excess risk

Theorem 2.4.1 (Crammer et al., 2008)

Under bounded loss with VC class $\mathcal{H}$ and posterior gap Δ , with high probability,

$$R^{(0)}(\hat{h}_{\mathbf{w}}) - \min_h R^{(0)}(h) \leq C w_1 \Delta + \tilde{O}\left(\sqrt{\text{VC}(\mathcal{H}) (w_0^2/n_0 + w_1^2/n_1)}\right).$$

Optimizing in $\mathbf{w}$:

$$w_1^* \propto \frac{n_1}{1+n_1\Delta^2}, \quad w_0^* \propto \frac{n_0}{1}.$$

E.g. $\Delta = 0.01, n_1 = 10^4$: $1 + n_1\Delta^2 = 2$, $w_1^ \approx n_1/2$ (half weight). $\Delta = 0.1$: $1 + n_1\Delta^2 = 101$, $w_1^* \approx n_1/101$ (down-weighted 100 $\times$).*

- Small Δ : source dominates (huge weight; $w_1^* \rightarrow n_1$).
- Large Δ : source weight collapses; we fall back to target-only.

The posterior-drift setting returns in L_4 , there handled by biased regularization.

§2.4: Hardness

Hardness: when transfer is provably impossible

We have matching upper bounds; now the lower bound — is the divergence term actually avoidable?

Consider covariate shift only ($f^{(0)} = f^{(1)} = f^*$). Even then:

Theorem 2.5.1 (Lower bound, informal; Ben-David et al., 2010b; Ben-David and Urner, 2012)

There exist source/target pairs $(\mathbb{P}^{(1)}, \mathbb{P}^{(0)})$ with $d_{\text{TV}}(\mathbb{P}^{(1)}, \mathbb{P}^{(0)}) \geq \tau$ such that no algorithm using only source samples achieves $R^{(0)}(\hat{h}) \leq \min_h R^{(0)}(h) + \tau/2$ in the worst case.

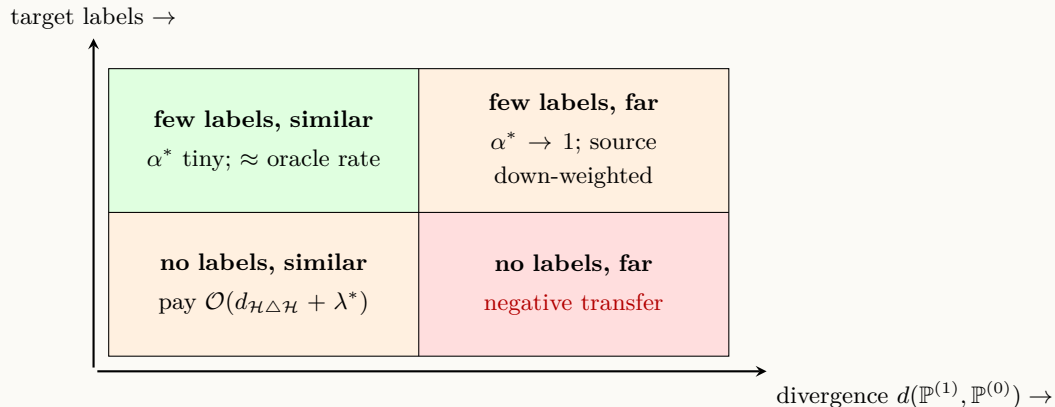
Hardness: what it implies

Implications:

- Without target labels, the divergence term in the upper bound *cannot be removed*.
- Lower bound matches upper bound up to constants $\Rightarrow$ TV captures the right notion of hardness for this setting.
- With target labels, even mild source-target similarity gets you back to oracle-rate up to a bias.

In plain terms: if source and target differ enough ($d_{\text{TV}} \geq \tau$), there are always two worlds — same source, different target — no source-only algorithm can tell apart. More source data never helps.

Hardness: the picture in one diagram



Refined later: density-ratio reweighting (L3) and biased regularization (L4).

Check-in: hardness

Two researchers compete: A uses 20,000 source labels with $d_{\mathcal{H}\Delta\mathcal{H}} = 0.4$; B uses 50 fresh target labels.

Who has the better worst-case bound on target excess risk, assuming $\text{VC}(\mathcal{H}) = 10$?

- (a) A, because $n_1 = 20,000$ is so much bigger than $n_0 = 50$.
- (b) B, because target sampling error $\sqrt{10/50} \approx 0.45$ beats the shift cost.
- (c) Both achieve the same rate; sample size is what matters.
- (d) Neither, because no algorithm beats the lower bound.

Answer

(b) – approximately tied with A. Researcher A pays ~ 0.4 (irreducible shift), while B pays ~ 0.45 (sample noise). But B's bound *vanishes* at $n_0 \rightarrow \infty$; A's does not. This is the hardness lower bound in action.

Check-in: hardness

Two researchers compete: A uses 20,000 source labels with $d_{\mathcal{H}\Delta\mathcal{H}} = 0.4$; B uses 50 fresh target labels.

Who has the better worst-case bound on target excess risk, assuming $\text{VC}(\mathcal{H}) = 10$?

- (a) A, because $n_1 = 20,000$ is so much bigger than $n_0 = 50$.
- (b) B, because target sampling error $\sqrt{10/50} \approx 0.45$ beats the shift cost.
- (c) Both achieve the same rate; sample size is what matters.
- (d) Neither, because no algorithm beats the lower bound.

Answer

(b) – approximately tied with A. Researcher A pays ~ 0.4 (irreducible shift), while B pays ~ 0.45 (sample noise). But B's bound *vanishes* at $n_0 \rightarrow \infty$; A's does not. This is the hardness lower bound in action.

§2.5: Beyond divergence

Other similarity notions

Divergence-style bounds have two systemic weaknesses:

- They can be *vacuous* when $d_{\mathcal{H}\Delta\mathcal{H}}$ is large but the joint task is still feasible.
- They depend only on *marginals* of X , ignoring conditional structure.

Three alternative similarity notions:

- λ^* joint-optimal error (Ben-David et al., 2010a); transfer exponents (Hanneke and Kpotufe, 2019): joint feasibility across two tasks.
- $\mathcal{A}$ -distance / proxy $\mathcal{A}$ -distance: estimate $d_{\mathcal{H}\Delta\mathcal{H}}$ from finite samples by training a binary classifier on X -domain labels.
- Integral probability metrics (e.g. MMD): kernel-based marginal distance, easy to estimate, independent of $\mathcal{H}$.

Joint-optimal error λ^* and transfer exponents

Definition 2.6.1 (ideal joint hypothesis, Ben-David et al., 2010a)

For loss ℓ and class $\mathcal{H}$,

$$\lambda^*(\mathbb{P}^{(0)}, \mathbb{P}^{(1)}; \mathcal{H}) = \min_{h \in \mathcal{H}} [R^{(0)}(h) + R^{(1)}(h)].$$

$\lambda^* > 0$ means no single $h \in \mathcal{H}$ fits both domains at once — irreducible misspecification, not a data problem.

- Captures *joint feasibility*: if some h is good on both, λ^* is small even if marginals are far apart.
- It is the irreducible bias term in the bounds above (e.g. the Blitzer bound).
- **Beyond divergence:** Hanneke and Kpotufe (2019) replace $\lambda^*/d_{\mathcal{H}\Delta\mathcal{H}}$ with *transfer exponents* (ρ, γ) , giving minimax rates in (n_0, n_1) achievable *adaptively*, without knowing the parameters.

Proxy $\mathcal{A}$ -distance and MMD

- **Proxy $\mathcal{A}$ -distance:** train a domain classifier (label X as source/target). Test error $\hat{\epsilon}$ gives $\widehat{d}_{\mathcal{A}} = 2(1 - 2\hat{\epsilon})$.
(random-guess domain classifier $\hat{\epsilon} = 0.5 \Rightarrow \hat{d}_{\mathcal{A}} = 0$; perfect separator $\hat{\epsilon} = 0 \Rightarrow \hat{d}_{\mathcal{A}} = 2$.)
 - ▷ Easy to compute; widely used as a *proxy* for $d_{\mathcal{H} \Delta \mathcal{H}}$.
- **Maximum Mean Discrepancy:** kernel embedding distance, $\text{MMD}^2 = \|\mu_{\mathbb{P}(0)} - \mu_{\mathbb{P}(1)}\|_{\mathcal{H}_K}^2$.
 - ▷ Doesn't depend on $\mathcal{H}$; well-defined for distributions; U -statistic estimator.
 - ▷ Powers many neural-network domain-adversarial losses (DANN, MMD-based UDA).

Lecture summary

Three escalating bounds for source-only learning:

$$\text{TV} \rightarrow d_{\mathcal{H} \Delta \mathcal{H}} \rightarrow \text{disc}_\ell.$$

Two qualitative regimes for source + target:

- Small shift, $\alpha^* \rightarrow n_0/(n_0 + n_1)$: pool everything.
- Large shift, $\alpha^* \rightarrow 1$: ignore source.

Hardness lower bounds say the divergence term is *not* an artifact of the proof.

Beyond divergence: λ^* , $\mathcal{A}$ -distance, MMD give complementary similarity measures.

Next: Lecture 3 takes the covariate-shift case seriously and replaces d_{TV} by an *importance-weighted* ERM that achieves oracle rates whenever the density ratio is well-behaved.

References

References

- Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., and Vaughan, J. W. (2010a). A theory of learning from different domains. *Machine learning*, 79:151–175.
- Ben-David, S., Blitzer, J., Crammer, K., and Pereira, F. (2006). Analysis of representations for domain adaptation. *Advances in neural information processing systems*, 19.
- Ben-David, S., Lu, T., Luu, T., and Pál, D. (2010b). Impossibility theorems for domain adaptation. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pages 129–136. JMLR Workshop and Conference Proceedings.
- Ben-David, S. and Urner, R. (2012). On the hardness of domain adaptation and the utility of unlabeled target samples. In *Algorithmic Learning Theory: 23rd International Conference, ALT 2012, Lyon, France, October 29-31, 2012. Proceedings 23*, pages 139–153. Springer.

- Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., and Wortman, J. (2007). Learning bounds for domain adaptation. *Advances in neural information processing systems*, 20.
- Crammer, K., Kearns, M., and Wortman, J. (2008). Learning from multiple sources. *Journal of Machine Learning Research*, 9(8).
- Hanneke, S. and Kpotufe, S. (2019). On the value of target data in transfer learning. *Advances in Neural Information Processing Systems*, 32.
- Kifer, D., Ben-David, S., and Gehrke, J. (2004). Detecting change in data streams. In *VLDB*, volume 4, pages 180–191. Toronto, Canada.
- Mansour, Y., Mohri, M., and Rostamizadeh, A. (2009). Domain adaptation: Learning bounds and algorithms. In *22nd Conference on Learning Theory, COLT 2009*.